

Higher Tier only questions

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Samples A, B and C contain magnesium or calcium ions ☒ Only samples A and B contain magnesium or calcium ions ☐ Only samples B and C contain magnesium or calcium ions ☐ None of the samples contain magnesium or calcium ions ☐			1	1		
	(b)			To show the expected result for permanent hard water ☐ To show the expected result for temporary hard water ☐ To act as a control for the experiment ☒ To make sure the investigation is a fair test ☐	1			1		1
	(c)			volume of soap solution and number of times shaken ☐ volume of water sample and volume of soap solution used ☐ volume of water sample and number of times shaken ☒ volume of water sample, volume of soap solution and number of times shaken ☐	1			1		1
	(d)			Sample B contains temporary hardness only ☐ Sample B contains permanent hardness only ☐ Sample B contains a mixture of temporary and permanent hardness ☒ Sample B does not contain hardness ☐ the volume of soap needed after boiling is less than before / less soap is needed after boiling (1) the volume of soap after boiling is still more than the distilled water (1)			3	3		3